

Multiple integrals

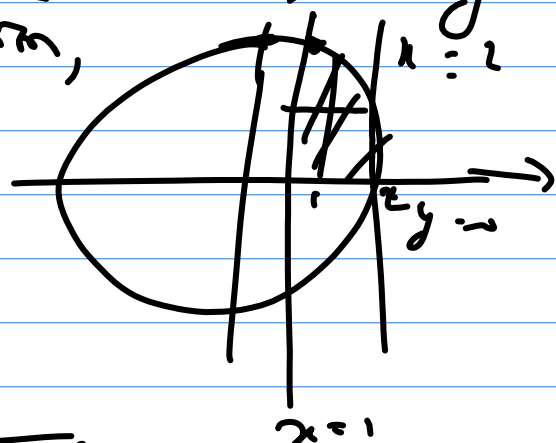
1) Evaluate: $\int_1^2 \int_0^{\sqrt{2-x^2}} \frac{x \, dy \, dx}{\sqrt{x^2+y^2}}$

Soln

Region: $0 \leq y \leq \sqrt{2-x^2} \quad \therefore y=0, y=\sqrt{2-x^2}$
 $1 \leq x \leq 2, \therefore x=1, x=2$

Changing order of integration,

$$= \int_0^1 \int_1^{\sqrt{2-y^2}} \frac{x \, dx \, dy}{\sqrt{x^2+y^2}}$$



$$= \frac{1}{2} \int_0^1 \int_1^{\sqrt{2-y^2}} \frac{2x \, dx \, dy}{\sqrt{x^2+y^2}}$$

$$= \frac{1}{2} \int_0^1 \left(\frac{(x^2+y^2)^{\frac{1}{2}}}{\frac{1}{x}} \right)_1^{\sqrt{2-y^2}} dy$$

$$= \int_0^1 \left[\sqrt{2-y^2+y^2} - \sqrt{1+y^2} \right] dy$$

$$= \int_0^1 \left[\sqrt{2} - \sqrt{1+y^2} \right] dy$$

$$= \left(\sqrt{2} y \right)_0^1 - \left(\frac{y}{2} \sqrt{1+y^2} + \frac{1}{2} \log(y + \sqrt{1+y^2}) \right)_0^1$$

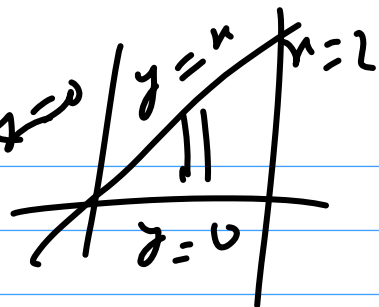
$$= \sqrt{2} - \left[\frac{1}{\sqrt{2}} + \frac{1}{2} \log(1+\sqrt{2}) - 0 - 0 \right]$$

$$= \sqrt{2} - \frac{1}{\sqrt{2}} - \frac{1}{2} \log(1+\sqrt{2})$$

$$= \frac{1}{\sqrt{2}} - \frac{1}{2} \log(1+\sqrt{2}) //$$

$$\int_0^2 \int_0^x y \, dy \, dx$$

Region: $0 \leq y \leq x$, $y=0$, $y=x$ ~~and~~
 $0 \leq x \leq 2$, $x=0$, $x=2$



changing to polar,

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dx \, dy = r \, dr \, d\theta$$

$$r = 0 \text{ to } r = 2$$

$$r \cos \theta = 2$$

$$r = 2 \sec \theta$$

$$\theta = 0 \text{ to } y = x$$

$$r \sin \theta = r \cos \theta$$

$$\theta = \frac{\pi}{4}$$

$$\int_0^{\pi/4} \int_0^{2 \sec \theta} r \sin \theta \cdot r \, dr \, d\theta$$

$$= \int_0^{\pi/4} \sin \theta \left[\frac{r^2}{2} \right]_0^{2 \sec \theta} d\theta$$

$$= \int_0^{\pi/4} \sin \theta \cdot \frac{8 \sec^2 \theta}{2} d\theta$$

$$= \frac{8}{2} \int_0^{\pi/4} \tan \theta \cdot \sec^2 \theta \, d\theta$$

$$= \frac{8}{2} \int_0^1 u \, du$$

$$= \frac{8}{2} \left(\frac{u^2}{2} \right)_0^1$$

$$= \frac{4}{2} //$$

$$\text{Let } u = \tan \theta$$

$$du = \sec^2 \theta \, d\theta$$

$$\text{When } \theta = 0, u = 0$$

$$\theta = \frac{\pi}{4}, u = 1$$

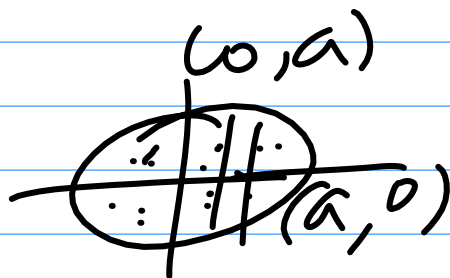
$$x^2 + y^2 = a^2$$

$$y = \pm \sqrt{a^2 - x^2}$$

$$-\sqrt{a^2 - x^2} \leq y \leq \sqrt{a^2 - x^2}$$

$$\text{Put } y = 0, x = \pm a$$

$$-a \leq x \leq a$$



$$we have, z = \frac{x^2 + y^2}{a}$$

$$Volume, = \iint z \, dA$$

$$= \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \frac{x^2 + y^2}{a} \, dy \, dx$$

changing into polar form,
 $0 \leq r \leq a$, $0 \leq \theta \leq 2\pi$

$$= \int_0^{2\pi} \int_0^a \left(\frac{r^2 \cos^2 \theta + r^2 \sin^2 \theta}{a} \right) r \, dr \, d\theta$$

$$= \frac{1}{a} \int_0^{2\pi} \int_0^a r^3 \, dr \, d\theta$$

$$= \frac{1}{a} \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^a \, d\theta$$

$$= \frac{1}{4} \int_0^{2\pi} a^3 \, d\theta$$

$$= \frac{a^3 \pi}{2}$$

$$x^2 + y^2 = 4$$

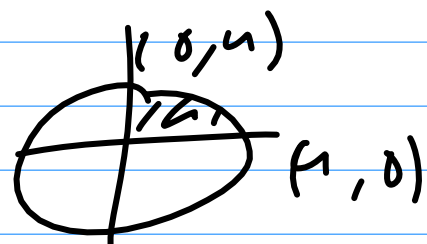
$$Region: 0 \leq y \leq \sqrt{4-x^2}$$

$$0 \leq x \leq 2$$

$$Volume, z = 3 - y$$

$$V = \iint_R z \, dA$$

in first octant
 plane: $z + y = 3$
 $z = 3 - y$



$$\begin{aligned}
 V &= \int_0^2 \int_0^{\sqrt{4-x^2}} (3-y) dy dx \\
 &= \int_0^{\frac{\pi}{2}} \int_0^2 (3r - r^2 \sin \theta) dr d\theta \quad \left\{ \begin{array}{l} \text{polar} \\ \text{conversion} \end{array} \right\} \\
 &= \int_0^{\frac{\pi}{2}} \left(\frac{3r^2}{2} - \frac{r^3}{3} \sin \theta \right) \Big|_0^2 d\theta \\
 &= \int_0^{\frac{\pi}{2}} \left(6 - \frac{8}{3} \sin \theta \right) d\theta \\
 &= \left[6\theta + \frac{8}{3} \cos \theta \right]_0^{\frac{\pi}{2}} \Rightarrow 3\pi + 0 - 0 - \frac{8}{3} \\
 &\quad \Rightarrow \underline{\underline{3\pi - \frac{8}{3}}}
 \end{aligned}$$

Asteroid

$$\begin{cases} x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \\ y = \left(a^{\frac{2}{3}} - x^{\frac{2}{3}} \right)^{\frac{3}{2}} \end{cases}$$

Dirichlet's theorem:

$x + y + z = 1$, then,

$$\iiint_V x^{a-1} y^{m-1} z^{n-1} dx dy dz = \frac{\Gamma(a) \cdot \Gamma(m) \cdot \Gamma(n)}{\Gamma(a+m+n+1)}$$

Q) $\iiint_V x^2 dx dy dz$, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Soln

$$\text{Put, } x = au, y = bv, z = cw$$

$$dx = a du, dy = b dv, dz = c dw$$

So, $u + v + w = 1$
Then,
$$\iiint a^2 u^2 \cdot a du \cdot b dv \cdot c dw$$

$$= a^3 b c \int \int \int u^2 du \cdot dv \cdot dw$$

$$= a^3 b c \left[\frac{\int_0^1 \int_0^{1-u} \int_0^{1-u-v} u^2 du \cdot dv \cdot dw}{\int_0^1 \int_0^{1-u} \int_0^{1-u-v} 1 du \cdot dv \cdot dw} \right] = a^3 b c \cdot \frac{2}{5!}$$

$$= \frac{a^3 b c}{60}$$

Q Find volume of ellipsoid, $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Soln

$$z = \pm \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$$

Put $z = 0$, $y = \pm b \sqrt{1 - \frac{x^2}{a^2}}$

Put $z = 0$, $y = 0$, $x = \pm a$

$$\text{Volume} = 8 \int_0^a \int_0^{b \sqrt{1 - \frac{x^2}{a^2}}} \int_0^{\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} dz dy dx$$

$$= 8 \int_0^a \int_0^{b \sqrt{1 - \frac{x^2}{a^2}}} c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} dy dx$$

Put, $\frac{p^2}{b^2} = 1 - \frac{x^2}{a^2}$

$$= 8 \int_0^a \int_0^p c \sqrt{\frac{p^2}{b^2} - \frac{y^2}{b^2}} dy dx = \frac{8c}{b} \int_0^a \int_0^p \sqrt{p^2 - y^2} dy dx$$

$$\begin{aligned}
&= \frac{8c}{b} \int_0^a \left[\frac{y\sqrt{p^2-y^2}}{2} + \frac{p^2}{2} \sin^{-1} \frac{y}{p} \right]_0^p dx \\
&= \frac{8c}{b} \int_0^a \left[0 + \frac{p^2 \pi}{4} \right] dx \\
&= \frac{8c\pi}{4b} \int_0^a b^2 \left(1 - \frac{x^2}{a^2} \right) dx \\
&= \frac{2bc\pi}{a^2} \int_0^a (a^2 - x^2) dx \\
&= \frac{2bc\pi}{a^2} \left[a^2 x - \frac{x^3}{3} \right]_0^a \\
&= \frac{2bc\pi}{a^2} \left[a^3 - \frac{a^3}{3} \right] \\
&= \frac{2bc\pi}{a^2} \times \frac{2a^3}{3} \\
&= \frac{4}{3} abc\pi
\end{aligned}$$

~~Gangad~~

Laplace transform

$$\rightarrow L(t^n \cdot f(t)) = (-1)^n \frac{d^n F(s)}{ds^n}$$

$$\rightarrow L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(s) ds$$

$$\rightarrow L\left(\int_0^t f(t) dt\right) = \frac{F(s)}{s}$$

Some formulas to remember

$$1) \sin A \cdot \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$2) \sin A \cdot \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$3) \cos A \cdot \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$1) \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$2) \cos 3x = 4 \cos^3 x - 3 \cos x$$

$$3) \sinh 3x = 3 \sinh x + 4 \sinh^3 x$$

$$4) \cosh 3x = 4 \cosh^3 x - 3 \cosh x$$

$$1) \cosh at = \frac{e^{at} + e^{-at}}{2}$$

$$2) \sinh at = \frac{e^{at} - e^{-at}}{2}$$

d) $\cosh at \cdot \sin at$

soln
Let $f(t) = \cosh at \cdot \sin at = \left(\frac{e^{at} + e^{-at}}{2} \right) \sin at$

$$L[f(t)] = \frac{1}{2} L[e^{at} \sin at + e^{-at} \sin at]$$

$$= \frac{1}{2} \left[\frac{a}{s^2 + a^2} \right]_{s \rightarrow s-a} + \frac{1}{2} \left[\frac{a}{s^2 + a^2} \right]_{s \rightarrow s+a}$$

$$= \frac{a}{2[(s-a)^2 + a^2]} + \frac{a}{2[(s+a)^2 + a^2]}$$

$$= \frac{a}{2} \left[\frac{1}{(s^2 + 2a^2) - 2as} + \frac{1}{(s^2 + 2a^2) + 2as} \right]$$

$$= \frac{a}{2} \left[\frac{(s^2 + 2a^2) + 2as + (s^2 + 2a^2) - 2as}{(s^2 + 2a^2)^2 - (2as)^2} \right]$$

$$= \frac{a(s^2 + 2a^2)}{s^4 + 4a^4} //$$

$$Q) e^{2t} \sin t \cdot \sin 2t \cdot \sin 3t = f(t)$$

$$f(t) = e^{2t} \cdot \frac{1}{2} [\cos(t-2t) - \cos(t+2t)] \cdot \sin 3t$$

$$= e^{2t} \frac{1}{2} [\cos t - \cos 3t] \cdot \sin 3t$$

$$= \frac{e^{2t}}{2} \left[\sin 3t \cdot \cos t - \frac{2}{2} \sin 3t \cos 3t \right]$$

$$= \frac{e^{2t}}{2} \left[\frac{1}{2} (\sin 4t + \sin 2t) - \frac{\sin 6t}{2} \right]$$

$$= \frac{e^{2t}}{4} [\sin 4t + \sin 2t - \sin 6t]$$

∴ From 1st shifting theorem,

$$\therefore \frac{1}{4} \mathcal{L} [e^{2t} \sin 4t + e^{2t} \sin 2t - e^{2t} \sin 6t]$$

$$= \frac{1}{4} \left\{ \left[\frac{4}{s^2+16} + \frac{2}{s^2+4} - \frac{6}{s^2+36} \right]_{s \rightarrow s-2} \right\}$$

$$= \frac{1}{4} \left[\frac{4}{(s-2)^2+16} + \frac{2}{(s-2)^2+4} - \frac{6}{(s-2)^2+36} \right] //$$

$$Q) t^2 \cos \omega t$$

$$\mathcal{L}(t^2 \cos \omega t) = (-1)^2 \frac{d^2}{ds^2} \left(\frac{s}{s^2+\omega^2} \right)$$

$$= \frac{d}{ds} \left(\frac{s^2+\omega^2 - s \cdot 2s}{(s^2+\omega^2)^2} \right) = \frac{d}{ds} \left(\frac{\omega^2 - s^2}{(s^2+\omega^2)^2} \right)$$

$$= \frac{(s^2+\omega^2)^2 (-2s) - (\omega^2 - s^2) 2(s^2+\omega^2) \cdot 2s}{(s^2+\omega^2)^4}$$

$$= \frac{-2s(s^2+\omega^2)^2 - 4s(s^2+\omega^2)(\omega^2 - s^2)}{(s^2+\omega^2)^4}$$

$$= \frac{-2s(s^2+\omega^2) - 4s(\omega^2 - s^2)}{(s^2+\omega^2)^3}$$

$$= \frac{-2s^3 - 2s\omega^2 - 4s\omega^2 + 4s^3}{(s^2+\omega^2)^3}$$

$$= \frac{2s^3 - 6s\omega^2}{(s^2 + \omega^2)^3} //$$

Q) $t e^{-3t} \cos 2t = f(t)$

Soln

Using 1st ... , $F(s) = \left(\frac{s}{s^2 + 4} \right)_{s \rightarrow s+3}$

$$F(s) = \left[\frac{s+3}{(s+3)^2 + 4} \right]$$

Now

$$L(t e^{-3t} \cos 2t) = (-1) \frac{d}{ds} F(s) = - \frac{d}{ds} \left[\frac{s+3}{(s+3)^2 + 4} \right]$$

$$= - \left[\frac{(s+3)^2 + 4 - (s+3)2(s+3)}{((s+3)^2 + 4)^2} \right]$$

$$= - \left[\frac{s^2 + 6s + 13 - 2(s+3)^2}{(s^2 + 6s + 13)} \right]$$

$$= - \left[\frac{s^2 + 6s + 13 - 2s^2 - 12s - 18}{s^2 + 6s + 13} \right]$$

$$= \frac{s^2 + 6s + 5}{s^2 + 6s + 13} //$$

Q) $\frac{\cos at - \cos bt}{t}$

Soln $L\left(\frac{\cos at - \cos bt}{t}\right) = \int_s^\infty L(\cos at - \cos bt) ds$

$$= \int_s^\infty \frac{s}{s^2 + a^2} ds - \int_s^\infty \frac{s}{s^2 + b^2} ds$$

$$= \left[\log(s^2 + a^2) \right]_s^\infty - \left[\log(s^2 + b^2) \right]_s^\infty$$

$$= \frac{1}{2} \left[\log\left(\frac{s^2 + a^2}{s^2 + b^2}\right) \right]_s^\infty = \frac{1}{2} \log \left[\frac{1 + \frac{a^2}{s^2}}{1 + \frac{b^2}{s^2}} \right]_s^\infty$$

$$\begin{aligned}
&= \frac{1}{2} \left[\log(1) - \log\left(\frac{s^2+a^2}{s^2+b^2}\right) \right] \\
&= -\frac{1}{2} \log\left(\frac{s^2+a^2}{s^2+b^2}\right) \\
&= \log \sqrt{\frac{s^2+b^2}{s^2+a^2}} //
\end{aligned}$$

$$8) \int_0^t \frac{1-\cos t}{t} dt$$

Soln

$$\mathcal{L}\left(\frac{1-\cos t}{t}\right) = \int_s^\infty \mathcal{L}(1-\cos t) dt = \int_s^\infty \left[\frac{1}{s} - \frac{s}{s^2+1}\right] ds$$

$$\Rightarrow \left[\log s - \frac{\log(s^2+1)}{2} \right]_s^\infty$$

$$= \left[\log\left(\frac{s}{(s^2+1)^{1/2}}\right) \right]_s^\infty$$

$$= \left[\log\left(\frac{1}{s(s^2+1)^{1/2}}\right) \right]_s^\infty$$

$$= \left[0 - \log \frac{s}{(s^2+1)^{1/2}} \right]$$

$$= \log \frac{(s^2+1)^{1/2}}{s}$$

$$\mathcal{L}\left(\int_0^t \frac{1-\cos t}{t} dt\right) = \frac{1}{s} \left(\log \frac{(s^2+1)^{1/2}}{s} \right) //$$

$$Q) F(s) = \frac{3s+2}{s^2+2s+2}$$

$$\underline{\text{Soln}} \quad f(t) = \mathcal{L}^{-1}\left(\frac{3s+3-1}{s^2+2s+2}\right)$$

$$= \mathcal{L}^{-1}\left[\frac{3(s+1)}{(s+1)^2+1} - \frac{1}{(s+1)^2+1}\right]$$

$$= \mathcal{L}^{-1}\left[3e^{-t} \left(\frac{s}{s^2+1}\right) - e^{-t} \left(\frac{1}{s^2+1}\right)\right]$$

$s \rightarrow s+1 \qquad \qquad \qquad s \rightarrow s+1$

$$= 3e^{-t} \cos t - e^{-t} \sin t //$$

Q) $\cot^{-1}\left(\frac{s}{\omega}\right)$

Let, $F(s) = \cot^{-1}\left(\frac{s}{\omega}\right)$

$$\left\{ \frac{d}{dx} \cot^{-1} x = -\frac{1}{1+x^2} \right\}$$

Obs wrt s , *

$$\frac{dF(s)}{ds} = -\frac{1}{1+\frac{s^2}{\omega^2}} \cdot \frac{1}{\omega} \Rightarrow -\frac{\omega}{s^2+\omega^2}$$

We know, $L\{t \cdot f(t)\} = -\frac{d}{ds}(F(s))$

$$t \cdot f(t) = L^{-1}\left(\frac{\omega}{s^2+\omega^2}\right) = \sin \omega t$$

$$f(t) = \frac{\sin \omega t}{t} //$$

2nd Shifting theorem,

$$L\{f(t-a) \cdot u(t-a)\} = e^{-as} F(s)$$

Note: $L[f(t) \cdot u(t-a)] = e^{-as} L[f(t+a)]$

Proof: By defⁿ of Laplace transform

$$L[f(t-a) \cdot u(t-a)] = \int_0^{\infty} e^{-st} f(t-a) \cdot u(t-a) dt$$

$$= \int_0^a e^{-st} f(t-a) \cdot 0 dt + \int_a^{\infty} e^{-st} f(t-a) \cdot dt$$

$$= \int_a^{\infty} e^{-st} f(t-a) dt$$

Put, $u = t - a$, when, $t = a$, $u = 0$
 $t = \infty$, $u = \infty$
 $du = dt$

$$= \int_0^{\infty} e^{-s(u+a)} f(u) du$$

$$= \int_0^{\infty} e^{-as} \cdot e^{-su} f(u) du$$

$$= e^{-as} F(s) //$$

$$\int e^{ax} \cos bx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

$$\int e^{ax} \sin bx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

Handwritten notes on lined paper:

8/2/20

Hint $\frac{1}{2}$

$\frac{1}{2}$

behaviour

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